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Firearm cartridge case

Abstract

A firearm cartridge case includes a short action cartridge case comprising a cartridge case main body portion and a cartridge case neck portion that are interconnected by a frusto-conical shoulder portion to form a bottleneck shape, a length L1 of the cartridge case between a base of a rebated rim and an interface between the cartridge case main body portion and the frusto-conical shoulder portion, a ratio of the length L1 of the cartridge case and a length of a projectile receivable within the cartridge case neck portion is between 3.5-0.9.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION[S] (1) This application claims the priority as a continuation-in-part of U.S. Nonprovisional patent application Ser. No. 17/938,728 (01674-SPR) filed Oct. 7, 2022 which claims priority to U.S. Provisional Patent Application No. 63/315,305 filed Mar. 1, 2022.

BACKGROUND

(1) This disclosure generally relates to small arms ammunition and more specifically to a centerfire cartridge case.

(2) Firearm cartridges are available in a wide range of variations for different applications. However, the variations are not infinite, and there are important constraints on cartridge dimensions. A typical rifle cartridge has a bottleneck shape with a slightly tapered main body

portion extending from a base or head end to a tapered frusto-conical shoulder that transitions to a nearly cylindrical neck to receive a projectile.

(3) While cartridges may exist in any theoretical dimension, typical cartridges are limited in their dimensions due to the characteristics of gun powder and structural limitations of firearm actions which define the chamber within which the cartridge is received. In addition, most cartridges are members of a family in which each member of the family is a variation from a common parent cartridge. While cartridges outside of established families with established case head dimensions are theoretically possible, there are economic barriers due to the large tooling cost for a new cartridge.

(4) Cartridge dimensions are also generally restricted to the dimensions of the actions of the firearms which are generally produced in but a few size categories. The action of a firearm typically includes an opening rearward of the chamber through which a cartridge is received that has a length that limits the cartridge overall length.

(5) As it may be disadvantageous to employ an action significantly longer than that for the desired cartridge, there are typically only a limited number of action lengths. A “magnum” length action accepts cartridges that are over 3.34 inches in length with most cartridges being about 3.6 inches long. The 375 H&H magnum is a well-known example. A “long/standard” length action accepts cartridges that are over 2.8 inches in length but under 3.34 inches. The 30-06 is a well-known example. A “short” length action accepts cartridges that are under 2.8 inches in length and generally any cartridge over about 2.3 inches. The 308 Winchester is a well-known example and the most popular short action cartridge on the market. A ‘mini’ length action accepts cartridges that are below 2.3 inches in length. The 5.56×45 NATO and its civilian equivalent, the 223 Remington is a well-known example.

(6) The width of the action opening also limits the diameter of the accepted cartridge. Typical actions are also provided in standard widths. A “super magnum” width action typically admits a cartridge with a maximum diameter of 0.580 inch. This is sized to closely accommodate the 378 and 460 Weatherby which are super magnum length belted cartridges, for instance. A “magnum” width action typically admits a cartridge with a maximum diameter of 0.538 inch. This is sized to closely accommodate the 350 Remington Magnum which is a short-action belted cartridge; the 6.5 mm Remington Magnum which is a short-action belted cartridge; the 300 Winchester Magnum which is a long-action belted cartridge; the 7 mm Remington Magnum which is a long-action belted cartridge; and the 375 Magnum which is a magnum-length-action belted cartridge. A “Long/standard” width action typically admits a cartridge with a maximum diameter of 0.471 inches. This is sized to closely accommodate the 308 Winchester, and the 30-06 Springfield cartridges, for instance.

(7) The shooting and hunting industry have traditionally been accustomed to choosing either a small standard cartridge or a large magnum cartridge. Cases with different levels of performance have long been sought in the industry.

SUMMARY

(8) A firearm cartridge case according to one disclosed non-limiting embodiment of the present disclosure includes a short action cartridge case comprising a cartridge case main body portion and a cartridge case neck portion that are interconnected by a frusto-conical shoulder portion to form a bottleneck shape, a length L1 of the cartridge case between a base of a rebated rim and an interface between the cartridge case main body portion and the frusto-conical shoulder portion, a ratio of the length L1 of the cartridge case and a length of a projectile receivable within the cartridge case neck portion is between 3.5-0.9.

(9) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile is a 22 caliber projectile having a projectile length of 0.550-1.260 inches.

(10) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the ratio of the length L1 of the cartridge case and the length of the projectile is between 3.3-

1.4.

(11) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile is a 24 caliber projectile having a projectile length of 0.685-1.502 inches.

(12) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.6-1.2.

(13) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile is a 25 caliber projectile having a projectile length of 0.505-1.416 inches.

(14) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the ratio of the length L1 of the cartridge case and the length of the projectile is between 3.5-1.3.

(15) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile is a 26 caliber projectile having a projectile length of 0.828-1.530 inches.

(16) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.1-1.1.

(17) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile is a 27 caliber projectile having a projectile length of 0.985-1.490 inches.

(18) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the ratio of the length L1 of the cartridge case and the length of the projectile is between 1.8-1.2.

(19) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile is a 28 caliber projectile having a projectile length of 0.870-1.692 inches.

(20) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.0-1.0.

(21) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile is a 30 caliber projectile having a projectile length of 0.614-1.876 inches.

(22) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.8-0.9.

(23) A firearm cartridge according to one disclosed non-limiting embodiment of the present disclosure includes a short action cartridge case comprising a cartridge case main body portion and a cartridge case neck portion that are interconnected by a frusto-conical shoulder portion to form a bottleneck shape, a rebated rim which defines a cartridge case rim diameter less than a cartridge case diameter; and a projectile at least partially within a mouth of the cartridge case neck portion to form a cartridge overall length that is less than 3.0 inches, wherein a ratio of a length L1 of the cartridge case and a length of a bullet receivable within the cartridge case neck portion is between 3.5-0.9.

(24) A further embodiment of any of the foregoing embodiments of the present disclosure includes that a ratio of the cartridge case main body portion diameter to the cartridge case rim diameter is between 1.03-1.10, and a ratio of an overall short action cartridge case length and the cartridge case main body portion diameter is 3.8-4.2.

(25) A further embodiment of any of the foregoing embodiments of the present disclosure includes that the projectile has a projectile length of 0.550-1.876 inches.

(26) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be appreciated that however the following description and drawings are intended to be exemplary in nature and non-limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various features will become apparent to those skilled in the art from the following detailed description of the disclosed non-limiting embodiment. The drawings that accompany the detailed description can be briefly described as follows:
- (2) FIG. 1 is a side view of an exemplary embodiment of a cartridge according to one disclosed non-limiting embodiment.
- (3) FIG. 2 is a side view of a cartridge case of the cartridge in FIG. 1.
- (4) FIG. 3 is an expanded side view of the head end of the cartridge case of the cartridge in FIG. 1.

DETAILED DESCRIPTION

- (5) FIG. 1 schematically illustrates a firearm cartridge **20** that has an elongated bottle shaped case **20A** capable of withstanding internal gas pressures of about 53,000 CUP, or 65,000 psi. In a firearm chamber, propellant burns over a period of time which produces increased pressure until it reaches a peak, then the pressure decreases until the projectile exits the muzzle wherein any remaining pressure does not contribute to projectile velocity. Cartridge performance is typically limited by a maximum allowable peak internal pressure in the chamber of the firearm.
- (6) The cartridge case **20A** has a head end defining a substantially circular base **22** with an annular rim **24** adjacent to an extractor groove **25**. In this embodiment, the annular rim **24** may be 0.049 inches thick and the extractor groove **25** may be 0.045 inches thick. An open end opposite the base **22** defines a mouth **26** to receive a projectile **20B**. In this embodiment, a 0.420 headspace gauge is at 1.656+/-0.003 inches.
- (7) The base **22** includes a centrally located primer pocket **27** that communicates via a flash hole **29** with an interior **30** of the cartridge case **20A** in which the powder charge **20C** (illustrated schematically) is contained. In this embodiment, the primer pocket **27** may receive a small rifle primer **20D** (illustrated schematically).
- (8) With reference to FIG. 2, the cartridge case **20A** includes a main body portion **28** that extends forward from the extractor groove **25** toward the mouth **26**, and a relatively smaller neck portion **36** that are interconnected by a frusto-conical shoulder portion **38** to form the bottleneck shaped cartridge case. The main body portion **28** and the smaller diameter neck portion **36** may not be perfectly cylindrical, but may be rather slightly frusto-conical, narrowing slightly in a direction away from the base **22**.
- (9) The frusto-conical shoulder portion **38** extends at an angle V with respect to the axis A of the cartridge **20**. In one embodiment, the angle V is 30 degrees.
- (10) The extractor groove **25** is formed at the rear of the main body portion **28** and has a diameter smaller than that of the rim **24** and the main body portion **28**. The extractor groove **25** forms a surface upon which an extractor engages the cartridge case **20A** to facilitate ejection of the cartridge case **20A** from the firearm. The extractor groove **25** is typically only present on a cartridge that is considered a rimless cartridge. In this embodiment, the rim **24** is a rebated rim. A rebated rim cartridge has a rim that is smaller in diameter than the base of the cartridge case, serving only for extraction purposes. Functionally essentially the same as a rimless case, the rebated rim provides some additional benefits when considered in conjunction with other cartridges such as allowing a firearm to be converted to fire a larger-than-normal cartridge, as most of a firearms' action (loading/extraction mechanism) need not be altered so long as the rim size is preserved.
- (11) In this embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.407 inches, and the main body portion **28** may have a cartridge case diameter, designated as D, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter D of the main body portion **28** of the cartridge case **20A** is measured at a location W (also

shown in FIG. 3) which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**. In one embodiment, a ratio of the cartridge case diameter D to the rim diameter R is between 1.03-1.10 and more specifically is 1.06. In contrast, a standard case may have a rim and a bolt end of the body with a diameter of 0.471 inches and a magnum case may have a rim with a diameter of 0.538 inches and a bolt end of the body with a diameter of 0.550 inches which is ratio of the cartridge case diameter to the rim diameter of 1.03.

(12) The overall length of the cartridge case **20A**, designated as L, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the cartridge case **20A**, designated as L, is 1.92 to 2.10 inches, and more specifically 2.035 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length L over such diameter D (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(13) The mouth **26** is the opening at the most forward part of the cartridge case **20A** within which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter M that is 0.255 inches in diameter with a thickness of 0.015. In this embodiment, the cartridge case **20A** includes a neck interface diameter N that is 0.285 inches in diameter at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38**. In this embodiment, the cartridge case **20A** also includes a body interface diameter B that is 0.482 inches in diameter at an interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(14) According to one disclosed embodiment, a length L1 of the cartridge case **20A**, between the base **22** and the interface **40** is 1.770 inches and the length L2 of the cartridge case **20** between the base **22** and the interface **42** is 1.602 inches.

(15) Commonly, there is approximately a 0.5 inch difference in cartridge length between a short action cartridge and a long action cartridge. That difference may be smaller when considering just the cartridge case rather than the entire cartridge. For example, a short action cartridge might have a cartridge case with a length of 2.05-2.10 inches whereas a long action cartridge might have a cartridge case with a length greater than 2.1 inches. A long action case is larger than a short action case, so more powder propellant may be contained in a long action case compared to a short action case. For example, a short action standard case may have 28 to 44 grains of gun powder. A short action magnum case may have 55 to 70 grains of gun powder. A long action standard case may have 55 to 67 grains of gun powder. A long action magnum case may have 55 to 67 grains of gun powder. This leaves a gap in grain count from 45 to 55 and therefore a gap in performance. The described amounts of gun powder may be most relevant to calibers from 22 caliber to 30 caliber. The cartridge case **20A** increase the propellant burn efficiency and uniformity with the end result being higher muzzle velocity with its attendant down-range advantages of reduced projectile drop, reduced wind deflection, and higher impact energy, etc. and increased shot-to-shot pressure/velocity consistency offering the potential for improved accuracy.

(16) According to one disclosed embodiment, the cartridge case **20A** does not have a radially protruding belt on the outside of the cartridge case such as that often utilized on a magnum cartridge. Magnum cases are cases that allow for a greater amount of propellant powder in comparison to what is considered a standard case. A magnum case may also have a cartridge case head with a greater diameter than a standard case. Given that a standard case is smaller in size relative to a magnum case, the amount of powder propellant which can be contained in a magnum case is greater than the amount of powder propellant which can be contained in a standard case, however, adding a belt as typical in a magnum case would require reductions in its diameter and capacity to make it compatible with a standard short action magazine. In addition, a belt would increase the difficulties attendant to chambering the cartridges and reduce cartridge capacity for a

short action.

(17) The cartridge case **20A** provides for a short action cartridge with a rebated rim, is not a magnum, and has a higher powder capacity than a standard case. Rebated rims heretofore have not been used in short action cases. The cartridge case **20A** allows for a cartridge with lower recoil, short action, and higher performance compared to a standard cartridge. The cartridge case **20A** as disclosed herein utilizes a short action case and a rebated rim. This unique combination allows for a greater amount of propellant powder to be contained in the cartridge case while not requiring the cartridge case to be a long action case. Long action cases are not usable in firearms that are designed to be used with short action cases.

(18) There is a significant range of projectiles within any given caliber and each has different flight characteristics. Generally, a heavier bullet of the same design and shape will outperform a lighter version at longer ranges. Longer bullets with greater length to diameter ratios can potentially be more accurate under certain conditions. The longer bullet can have a higher ballistic coefficient, which can retain velocity and resist wind drift better than a shorter bullet. As length increases, sectional density, which is a predictor of penetration power in bullets, increases in direct relationship to weight and if the shape of the nose and tail remain constant, the ballistic coefficient increases as well. Ballistic coefficient is a numerical representation of a bullet's resistance to drag/aerodynamic forces due to wind. Although the heavier bullet will leave the muzzle of the rifle at a lower speed, the heavier bullet will retain that speed better than a lighter bullet and drop less at longer ranges as it will decelerate more slowly.

(19) The following are examples of the cartridge in various calibers, e.g., 22, 24, 25, 26, 27, 28 and 30. The dimensions unless recited otherwise, may follow those dimensions described above. Although particular example calibers are provided, it should be appreciated that other calibers will benefit herefrom.

(20) In one disclosed embodiment of a cartridge case **20A** referred to herein as 22 Sprenger, a 22 caliber projectile **20B** having a projectile length P of 0.550-1.260 inches may be utilized.

(21) In this 22 Sprenger embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.409 inches, and the main body portion **28** may have a cartridge case diameter, designated as D, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter D of the main body portion **28** of the cartridge case **20A** is measured at a location W which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**.

(22) The mouth **26** is the opening at the most forward part of the cartridge case **20A** through which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter M that is 0.222 inches in diameter. In this embodiment, the cartridge case **20A** includes the neck interface diameter N that is 0.252 inches in diameter measured at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38** such that the projectile **20B** with a projectile outer diameter ODP of 0.224 inches forms a neck outer diameter ODN of 0.254 inches after the projectile is seated therein. In this embodiment, the cartridge case **20A** also includes the body interface diameter B that is 0.482 inches in diameter at the interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(23) The overall length of the cartridge case **20A**, designated as L, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the cartridge case **20A**, designated as L, is 1.92 to 2.10 inches and more specifically 2.055 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length L over such diameter D (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(24) In this 22 Sprenger embodiment, the length L1 of the cartridge case **20A**, between the base **22**

and the interface **40** is 1.800 inches and the length **L2** of the cartridge case **20** between the base **22** and the interface **42** is 1.601+/-0.010 inches. A ratio of the length **L1** of the cartridge case and a projectile length **PA** of a 22 caliber projectile **20B** receivable within the cartridge case neck portion in this 22 caliber embodiment may be between 3.5-0.9, and more specifically may be between 3.3-1.4, and still more specifically may be 2.0.

(25) In another disclosed embodiment of a cartridge case **20A** referred to herein as 24 Sprenger, a 24 caliber projectile **20B** having a projectile length **PB** of 0.685-1.502 inches may be utilized.

(26) In this 24 Sprenger embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.409 inches, and the main body portion **28** may have a cartridge case diameter, designated as **D**, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter **D** of the main body portion **28** of the cartridge case **20A** is measured at a location **W** which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**.

(27) The mouth **26** is the opening at the most forward part of the cartridge case **20A** through which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter **M** that is 0.241 inches in diameter. In this embodiment, the cartridge case **20A** includes the neck interface diameter **N** that is 0.271 inches in diameter measured at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38** such that the projectile **20B** with a projectile outer diameter **ODP** of 0.243 inches forms a neck outer diameter **ODN** of 0.273 inches after the projectile is seated therein. In this embodiment, the cartridge case **20A** also includes the body interface diameter **B** that is 0.482 inches in diameter at the interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(28) The overall length of the cartridge case **20A**, designated as **L**, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the cartridge case **20A**, designated as **L**, is 1.92 to 2.10 inches, and more specifically 2.055 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length **L** over such diameter **D** (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(29) In this 24 Sprenger embodiment, the length **L1** of the cartridge case **20A**, between the base **22** and the interface **40** is 1.785 inches and the length **L2** of the cartridge case **20A** between the base **22** and the interface **42** is 1.602+/-0.010 inches. A ratio of the length **L1** of the cartridge case and a projectile length **PB** of a 24 caliber projectile **20B** receivable within the cartridge case neck portion in this 24 caliber embodiment may be between 3.5-0.9, and more specifically may be between 2.6-1.2, and still more specifically may be 1.6.

(30) In another disclosed embodiment of a cartridge case **20A** referred to herein as 25 Sprenger, a 25 caliber projectile **20B** having a projectile length **PA** of 0.505-1.416 inches may be utilized.

(31) In this 25 Sprenger embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.409 inches, and the main body portion **28** may have a cartridge case diameter, designated as **D**, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter **D** of the main body portion **28** of the cartridge case **20A** is measured at a location **W** which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**.

(32) The mouth **26** is the opening at the most forward part of the cartridge case **20A** through which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter **M** that is 0.255 inches in diameter. In this embodiment, the cartridge case **20A** includes the neck interface diameter **N** that is 0.285 inches in diameter measured at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38** such that the projectile **20B** with a projectile outer diameter **ODP**

of 0.257 inches forms a neck outer diameter ODN of 0.287 inches after the projectile is seated therein. In this embodiment, the cartridge case **20A** also includes the body interface diameter B that is 0.482 inches in diameter at the interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(33) The overall length of the cartridge case **20A**, designated as L, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the cartridge case **20A**, designated as L, is 1.92 to 2.10 inches and more specifically 2.045 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length L over such diameter D (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(34) In this 25 Sprenger embodiment, the length L1 of the cartridge case **20A**, between the base **22** and the interface **40** is 1.770 inches and the length L2 of the cartridge case **20** between the base **22** and the interface **42** is 1.602+/-0.010 inches. A ratio of the length L1 of the cartridge case and a projectile length PC of a 25 caliber projectile **20B** receivable within the cartridge case neck portion in this 25 caliber embodiment may be between 3.5-0.9, and more specifically may be between 3.5-1.3, and still more specifically may be 1.8.

(35) In another disclosed embodiment of a cartridge case **20A** referred to herein as 26 Sprenger, a 26 caliber projectile **20B** having a projectile length PA of 0.828-1.530 inches may be utilized.

(36) In this 26 Sprenger embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.409 inches, and the main body portion **28** may have a cartridge case diameter, designated as D, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter D of the main body portion **28** of the cartridge case **20A** is measured at a location W which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**.

(37) The mouth **26** is the opening at the most forward part of the cartridge case **20A** through which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter M that is 0.262 inches in diameter. In this embodiment, the cartridge case **20A** includes the neck interface diameter N that is 0.292 inches in diameter measured at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38** such that the projectile **20B** with a projectile outer diameter ODP of 0.264 inches forms a neck outer diameter ODN of 0.294 inches after the projectile is seated therein. In this embodiment, the cartridge case **20A** also includes the body interface diameter B that is 0.482 inches in diameter at the interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(38) The overall length of the cartridge case **20A**, designated as L, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the cartridge case **20A**, designated as L, is 1.92 to 2.10 inches and more specifically 2.035 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length L over such diameter D (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(39) In this 26 Sprenger embodiment, the length L1 of the cartridge case **20A**, between the base **22** and the interface **40** is 1.755 inches and the length L2 of the cartridge case **20A** between the base **22** and the interface **42** is 1.590+/-0.010 inches. A ratio of the length L1 of the cartridge case and a projectile length PD of a 26 caliber projectile **20B** receivable within the cartridge case neck portion in this 26 caliber embodiment may be between 3.5-0.9, and more specifically may be between 2.1-1.1 and still more specifically may be 1.5.

(40) In another disclosed embodiment of a cartridge case **20A** referred to herein as 27 Sprenger, may utilize a 27 caliber projectile **20B** having a projectile length PA of 0.985-1.490 inches.

(41) In this 27 Sprenger embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.409 inches, and the main body portion **28** may have a cartridge case diameter, designated as D, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter D of the main body portion **28** of the cartridge case **20A** is measured at a location W which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**.

(42) The mouth **26** is the opening at the most forward part of the cartridge case **20A** through which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter M that is 0.275 inches in diameter. In this embodiment, the cartridge case **20A** includes the neck interface diameter N that is 0.305 inches in diameter measured at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38** such that the projectile **20B** with a projectile outer diameter ODP of 0.277 inches forms a neck outer diameter ODN of 0.307 inches after the projectile is seated therein. In this embodiment, the cartridge case **20A** also includes the body interface diameter B that is 0.482 inches in diameter at the interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(43) The overall length of the cartridge case **20A**, designated as L, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the cartridge case **20A**, designated as L, is 1.92 to 2.10 inches and more specifically 2.035 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length L over such diameter D (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(44) In this 27 Sprenger embodiment, the length L1 of the cartridge case **20A**, between the base **22** and the interface **40** is 1.740 inches and the length L2 of the cartridge case **20A** between the base **22** and the interface **42** is 1.587+/-0.010 inches. A ratio of the length L1 of the cartridge case and a projectile length PE of a 27 caliber projectile **20E** receivable within the cartridge case neck portion in this 27 caliber embodiment may be between 3.5-0.9, and more specifically may be between 1.8-1.2 and still more specifically may be 1.4.

(45) In another disclosed embodiment of a cartridge case **20A** referred to herein as 28 Sprenger, a 28 caliber projectile **20B** having a projectile length PA of 0.870-1.692 inches may be utilized.

(46) In this 28 Sprenger embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.409 inches, and the main body portion **28** may have a cartridge case diameter, designated as D, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter D of the main body portion **28** of the cartridge case **20A** is measured at a location W which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**.

(47) The mouth **26** is the opening at the most forward part of the cartridge case **20A** through which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter M that is 0.282 inches in diameter. In this embodiment, the cartridge case **20A** includes the neck interface diameter N that is 0.312 inches in diameter measured at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38** such that the projectile **20B** with a projectile outer diameter ODP of 0.284 inches forms a neck outer diameter ODN of 0.314 inches after the projectile is seated therein. In this embodiment, the cartridge case **20A** also includes the body interface diameter B that is 0.482 inches in diameter at the interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(48) The overall length of the cartridge case **20A** in this 28 Sprenger embodiment, designated as L, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the

cartridge case **20A**, designated as L, is 1.92 to 2.10 inches and more specifically 2.035 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length L over such diameter D (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(49) In this 28 Sprenger embodiment, the length L1 of the cartridge case **20A**, between the base **22** and the interface **40** is 1.725 inches and the length L2 of the cartridge case **20A** between the base **22** and the interface **42** is 1.578+/-0.010 inches. A ratio of the length L1 of the cartridge case and a projectile length PF of a 28 caliber projectile **20F** receivable within the cartridge case neck portion in this 28 caliber embodiment may be between 3.5-0.9, and more specifically may be between 2.0-1.0 and still more specifically may be 1.3.

(50) In another disclosed embodiment of a cartridge case **20A** referred to herein as 30 Sprenger, a 30 caliber projectile **20B** having a projectile length PA of 0.614-1.876 inches may be utilized.

(51) In this 30 Sprenger embodiment, the rim **24** may have a diameter of 0.471 inches, the extractor groove **25** may have a diameter of 0.409 inches, and the main body portion **28** may have a cartridge case diameter, designated as D, of 0.497 inches. For purposes of that disclosed herein, the cartridge case diameter D of the main body portion **28** of the cartridge case **20A** is measured at a location W which is 0.150 inches from the base **22**, so as to identify such outer diameter precisely despite the slightly frusto-conical shape of the main body portion **28**.

(52) The mouth **26** is the opening at the most forward part of the cartridge case **20A** through which the projectile **20B** is seated. The projectile **20B** may be configured in many different sizes and shapes. In this embodiment, the cartridge case **20A** includes a mouth diameter M that is 0.306 inches in diameter. In this embodiment, the cartridge case **20A** includes the neck interface diameter N that is 0.336 inches in diameter measured at an interface **40** between the neck portion **36** and the frusto-conical shoulder portion **38** such that the projectile **20B** with a projectile outer diameter ODP of 0.308 inches forms a neck outer diameter ODN of 0.338 inches after the projectile is seated therein. In this embodiment, the cartridge case **20A** also includes the body interface diameter B that is 0.482 inches in diameter at the interface **42** of the frusto-conical shoulder portion **38** and the main body portion **28**.

(53) The overall length of the cartridge case **20A** in this 30 Sprenger embodiment, designated as L, extends between the base **22** at the head end of the cartridge case **20A** and the mouth **26** at the open end of the cartridge case **20A**. According to one disclosed embodiment, the overall length of the cartridge case **20A**, designated as L, is 1.92 to 2.10 inches and more specifically 2.035 inches. To maximize the powder-carrying capacity of the cartridge case **20A** in a manner which nevertheless produces a cartridge short enough to be used in a short-action firearm, the ratio of the overall case length L over such diameter D (i.e., L/D) should be no more than about 3.8-4.2, and more specifically 4.0.

(54) In this 30 Sprenger embodiment, the length L1 of the cartridge case **20A**, between the base **22** and the interface **40** is 1.710 inches and the length L2 of the cartridge case **20A** between the base **22** and the interface **42** is 1.584+/-0.010 inches. A ratio of the length L1 of the cartridge case and a projectile length PG of a 30 caliber projectile **20G** receivable within the cartridge case neck portion in this 30 caliber embodiment may be between 3.5-0.9, and more specifically may be between 2.8-0.9 and still more specifically may be 1.4.

(55) The cartridge case **20A** advantageously bridges a gap in performance that has not been previously identified. The cartridge case **20A** as disclosed herein, creates a performance window for high ballistic coefficient projectiles from a short action non-magnum case with greater propellant powder capacity.

(56) Although the different non-limiting embodiments have specific illustrated components, the embodiments of this invention are not limited to those particular combinations. It is possible to use some of the components or features from any of the non-limiting embodiments in combination with

features or components from any of the other non-limiting embodiments.

(57) The foregoing description is exemplary rather than defined by the limitations within. Various non-limiting embodiments are disclosed herein, however, one of ordinary skill in the art would recognize that various modifications and variations in light of the above teachings will fall within the scope of the appended claims. It is therefore to be appreciated that within the scope of the appended claims, the disclosure may be practiced other than as specifically described. For that reason, the appended claims should be studied to determine true scope and content.

Claims

1. A firearm cartridge case, comprising: a short action cartridge case comprising a cartridge case main body portion and a cartridge case neck portion that are interconnected by a frusto-conical shoulder portion at an interface to form a bottleneck shape, a length L1 of the cartridge case between a base of a rebated rim and the interface between the cartridge case main body portion and the frusto-conical shoulder portion, a ratio of the length L1 of the cartridge case and a length of a projectile receivable within the cartridge case neck portion is between 3.5-0.9.
2. The firearm cartridge case as recited in claim 1, wherein the projectile is a 22 caliber projectile having a projectile length of 0.550-1.260 inches.
3. The firearm cartridge case as recited in claim 2, wherein the ratio of the length L1 of the cartridge case and the length of the projectile is between 3.3-1.4.
4. The firearm cartridge case as recited in claim 1, wherein the projectile is a 24 caliber projectile having a projectile length of 0.685-1.502 inches.
5. The firearm cartridge case as recited in claim 4, wherein the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.6-1.2.
6. The firearm cartridge case as recited in claim 1, wherein the projectile is a 25 caliber projectile having a projectile length of 0.505-1.416 inches.
7. The firearm cartridge case as recited in claim 6, wherein the ratio of the length L1 of the cartridge case and the length of the projectile is between 3.5-1.3.
8. The firearm cartridge case as recited in claim 1, wherein the projectile is a 26 caliber projectile having a projectile length of 0.828-1.530 inches.
9. The firearm cartridge case as recited in claim 8, wherein the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.1-1.1.
10. The firearm cartridge case as recited in claim 1, wherein the projectile is a 27 caliber projectile having a projectile length of 0.985-1.490 inches.
11. The firearm cartridge case as recited in claim 10, wherein the ratio of the length L1 of the cartridge case and the length of the projectile is between 1.8-1.2.
12. The firearm cartridge case as recited in claim 1, wherein the projectile is a 28 caliber projectile having a projectile length of 0.870-1.692 inches.
13. The firearm cartridge case as recited in claim 12, wherein the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.0-1.0.
14. The firearm cartridge case as recited in claim 1, wherein the projectile is a 30 caliber projectile having a projectile length of 0.614-1.876 inches.
15. The firearm cartridge case as recited in claim 14, wherein the ratio of the length L1 of the cartridge case and the length of the projectile is between 2.8-0.9.
16. A firearm cartridge, comprising: a short action cartridge case comprising a cartridge case main body portion and a cartridge case neck portion that are interconnected by a frusto-conical shoulder portion at an interface to form a bottleneck shape, a rebated rim which defines a cartridge case rim diameter less than a cartridge case diameter, a length L1 of the cartridge case between a base of a rebated rim and the interface between the cartridge case main body portion and the frusto-conical shoulder portion; and a projectile at least partially within a mouth of the cartridge case neck portion

to form a cartridge overall length that is less than 3.0 inches, wherein a ratio of the length L1 of the cartridge case and a length of the projectile receivable within the cartridge case neck portion is between 3.5-0.9.

17. The firearm cartridge as recited in claim 16, wherein a ratio of the cartridge case main body portion diameter to the cartridge case rim diameter is between 1.03-1.10, and a ratio of an overall short action cartridge case length and the cartridge case main body portion diameter is 3.8-4.2.

18. The firearm cartridge as recited in claim 17, wherein the projectile has a projectile length of 0.550-1.876 inches.

19. The firearm cartridge case as recited in claim 18, wherein the main body portion extends forward from an extractor groove to the frusto-conical shoulder.

20. The firearm cartridge case as recited in claim 19, wherein the cartridge case main body portion diameter is 0.497 inches measured at a location which is 0.150 inches from the base.

21. The firearm cartridge case as recited in claim 20, wherein the cartridge case main body portion diameter is 0.497 inches measured at a location which is 0.150 inches from the base and the rim diameter is 0.471 inches.

22. The firearm cartridge case as recited in claim 21, wherein the short action cartridge case is 1.92 to 2.10 inches in length.

23. The firearm cartridge case as recited in claim 16, wherein a ratio of the cartridge case main body portion diameter to the cartridge case rim diameter is between 1.03-1.10.

24. The firearm cartridge case as recited in claim 16, wherein a length of the cartridge case between a base and an interface between the frusto-conical shoulder portion and the cartridge case neck portion is 1.770 inches and the length of the cartridge case between the base and an interface between the frusto-conical shoulder portion and the cartridge case main body portion is 1.602 inches.

25. The firearm cartridge case as recited in claim 24, wherein an angle defined by the frusto-conical shoulder portion is 30 degrees.
